

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR10074 COMPUTER COMMUNICATIONS AND NETWORKS

Monday 5th May 2025

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

1. Answer QUESTION 1 and ONE other question.
2. Question 1 is COMPULSORY. If both QUESTION 2 and QUESTION 3 are answered, only QUESTION 2 will be marked.
3. All questions carry equal weight.
4. This is a NOTES PERMITTED, CALCULATORS NOT PERMITTED examination.

Candidates may consult up to THREE A4 pages (6 sides) of notes.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION.

Year 3 Courses

Convener: P.Series

External Examiners: D.Parker, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. THIS QUESTION IS COMPULSORY

- (a) Explain why recursive DNS lookup is more vulnerable to DNS cache poisoning attacks than iterative DNS lookup. (maximum 80 words) [4 marks]
- (b) A layer 7 proxy splits a TCP connection into two, one to the client and the other to the server, and relays data between them. It could improve the end-to-end throughput when placed in a long RTT client-server path. Explain a possible reason. (maximum 70 words) [5 marks]
- (c) Consider implementing a middlebox that removes all the TCP options from every TCP packet that traverses it. All the traffic is carried over IPv4 and no tunneling is used. What does this middlebox have to do? Describe all the IPv4 and TCP header fields that need to be modified so that the resulting packets look correct at the receiver. [5 marks]
- (d) Assume a Bluetooth classic connection has been established at 1Mb/s. A data frame contains 72 bits of access code, 54 bits of header and 16 bits of CRC.
- i. Suppose the slave device wants to send data packets of 340 bytes to the master. If no forward error correction (FEC) is employed, how much time will be required to complete the transfer? [3 marks]
 - ii. Suppose 1/3 FEC for the payload is used. Calculate the impact this has on the effective data transfer rate for the link (assuming no retransmissions are needed). [3 marks]
- (e) In a partial packet trace of a TCP connection below:

```
09:43:46.176746 IP dhcp-90-159.inf.ed.ac.uk.45104 > 128.243.28.128.https:
  Flags [P.], seq 1801:1865, ack 239, win 501,
  options [nop,nop,TS val 2358845523 ecr 2215054993], length 64
09:43:46.176953 IP dhcp-90-159.inf.ed.ac.uk.45104 > 128.243.28.128.https:
  Flags [P.], seq 1865:1957, ack 239, win 501, options
  [nop,nop,TS val 2358845523 ecr 2215054993], length 92
09:43:46.177188 IP dhcp-90-159.inf.ed.ac.uk.45104 > 128.243.28.128.https:
  Flags [P.], seq 1957:2415, ack 239, win 501, options
  [nop,nop,TS val 2358845524 ecr 2215054993], length 458
09:43:46.187701 IP 128.243.28.128.https > dhcp-90-159.inf.ed.ac.uk.45104:
  Flags [.], ack 1957, win 249, options
  [nop,nop,TS val 2215055005 ecr 2358845523], length 0
```

- i. How many bytes of the application data have been transferred? [2 marks]
- ii. Which packets of the trace does the packet observed at 09:43:46.187701 acknowledge the reception? List all the packets applicable using the timestamp as the packet identity. [3 marks]

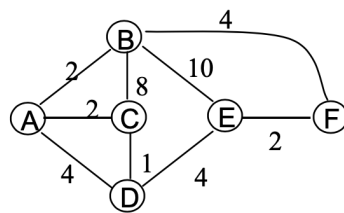
2. ANSWER EITHER THIS QUESTION OR QUESTION 3

- (a) Suppose a host has a 4-MB file that is to be sent to another host over a 3.2-Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress the data by 45%, or 2 seconds to compress by 50%. Ignore queueing delays, and compression and transmission cannot overlap. Calculate the total time of compression time, transmission delay and propagation delay for each of the compression options. [4 marks]
- (b) How can the routes below be represented as a single routing table entry after route aggregation?

Destination	Output interface
133.27.170.0/24	1
133.27.171.0/24	1

[3 marks]

- (c) Using Dijkstra's algorithm, compute the shortest path from E to all network nodes in the following network topology, and fill the possible cells in the table. Note that $D(v)$ denotes the cost of the least-cost path from the source node to destination v as of this iteration of the algorithm. $p(v)$ denotes the previous node (neighbor of v) along the current least-cost path from the source to v . N' is subset of nodes; v is in N' if the least-cost path from the source to v is definitively known. [6 marks]



N'	$D(A), p(A)$	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	$D(F), p(F)$
E	infinity	10,E			
EF					
EFD					
EFDC					
EFDCB					

- (d) Answer the questions about DCTCP below:
- Why does DCTCP achieve low latency of small messages? [2 marks]
 - Why does DCTCP achieve high throughput of large messages? [2 marks]
- (e) Explain how third-party CDNs redirect client requests into their network. (maximum 60 words) [5 marks]
- (f) Explain why TCP segmentation offload reduces the CPU usage of the host TCP stack (maximum 50 words). [3 marks]

3. ANSWER EITHER THIS QUESTION OR QUESTION 2

- (a) Assume that a TCP sender has sent 10 packets. Even when a single packet drop occurs, the TCP sender might have to wait for the expiration of retransmission timeout. When does this situation happen? [4 marks]
- (b) Using the AIMD algorithm, when RTT between the TCP sender and receiver is 200ms and the bandwidth is 1Gbps, how long does it take to recover the congestion window to the full amount after a packet loss when the maximum segment size is 1500 bytes? [5 marks]
- (c) Consider a link that transfers approximately $1/p$ consecutive packets followed by a single packet drop. In a sender that employs the AIMD algorithm, the congestion window will traverse a periodic sawtooth, where the maximum and minimum window sizes will be W and $W/2$, respectively. We want to derive the additive increase parameter a that achieves 4 times higher throughput than regular TCP where a is the number of packets by which the window size increases every RTT (regular TCP thus uses $a = 1$). Note that the multiplicative decrease parameter is the same with TCP (i.e., $1/2$). Ignore link utilization of ack packets.
- i. Define the amount of data sent in a single cycle, where the window increases from $W/2$ to W , in terms of W and a . [2 marks]
 - ii. Define W in terms of a and p . [1 mark]
 - iii. Define the throughput T in terms of a , p , the packet size S (assume all the packets have the same size), RTT R and W . [1 mark]
 - iv. From the result obtained in 1(b)iii, derive a that achieves 4 times higher throughput of regular TCP. [2 marks]
- (d) Unlike QUIC, TCP could allow NAT implementations to remove a translation table entry without relying on timeouts. Explain why. (maximum 50 words) [6 marks]
- (e) Assume that a network administrator would like to request the ISP to allocate an IPv4 address block (subnet). They need to accommodate 1900 hosts at the same time. What is the longest subnet length that supports this? Also explain why. [4 marks]